How to test the FloMo idea

Q) Given $\overset{\text{algorithm}}{A}$ that yields policy

$$\pi(a_{1:T}, o^f_{1:T} \mid o^s, \underset{\text{language-command/goal...}}{g})$$

I want to understand how noisier GT point tracks yield affect downstream performance

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What are the building blocks for such an experiment

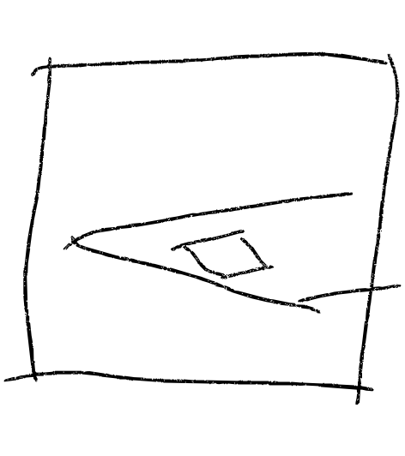
Single setup but goal-conditioned environment w/ lots of BC data OR a teacher policy that can generate this data reasonably quickly (at least real-time)

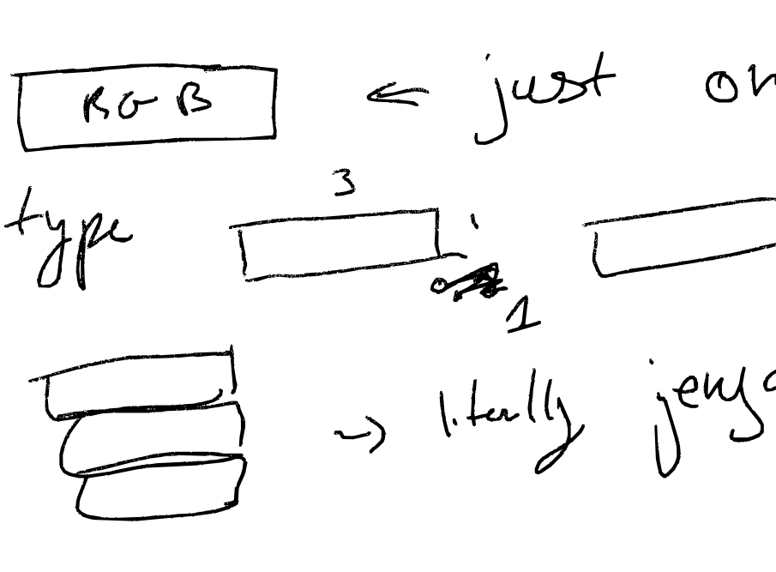
↳ I should look into the diversity of laptop / camera setups in Mani-Skill. It could be the right place...
  ↳ train flow prediction policy for action chunk length
  ↳ measure sample-complexity of no flow vs. noisy flow vs. gt flow.

Anticipated difficulties
- training policies
- implementing flo-mo policy.

What do I need to be confident about for a strong flo-mo policy?

How to specify this further?





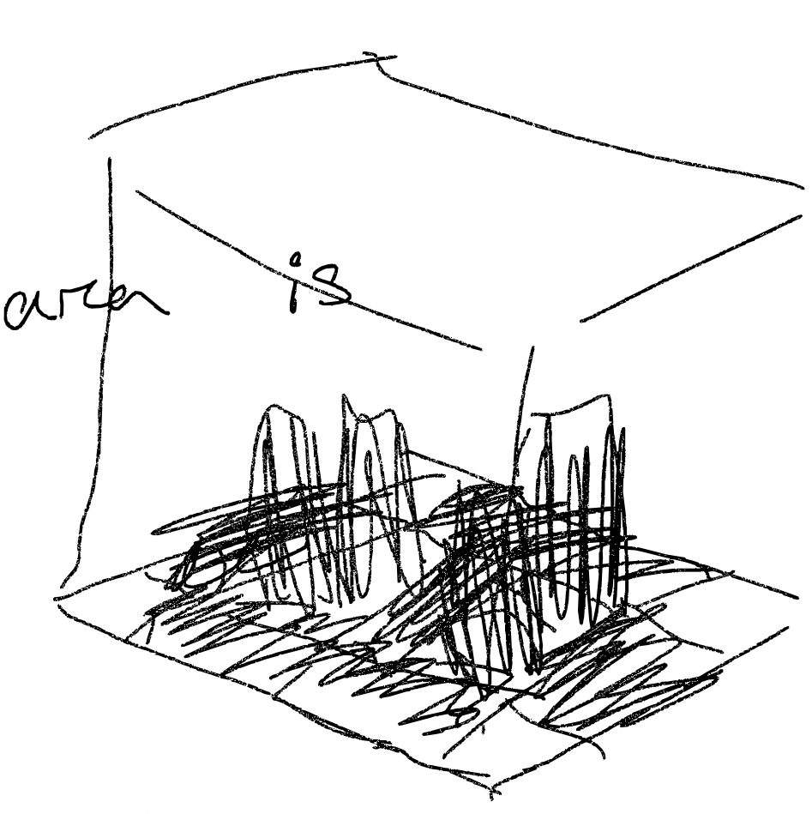
← just one block type

→ literally jenga structures

Jenga will be easy for CLAUDE to understand.

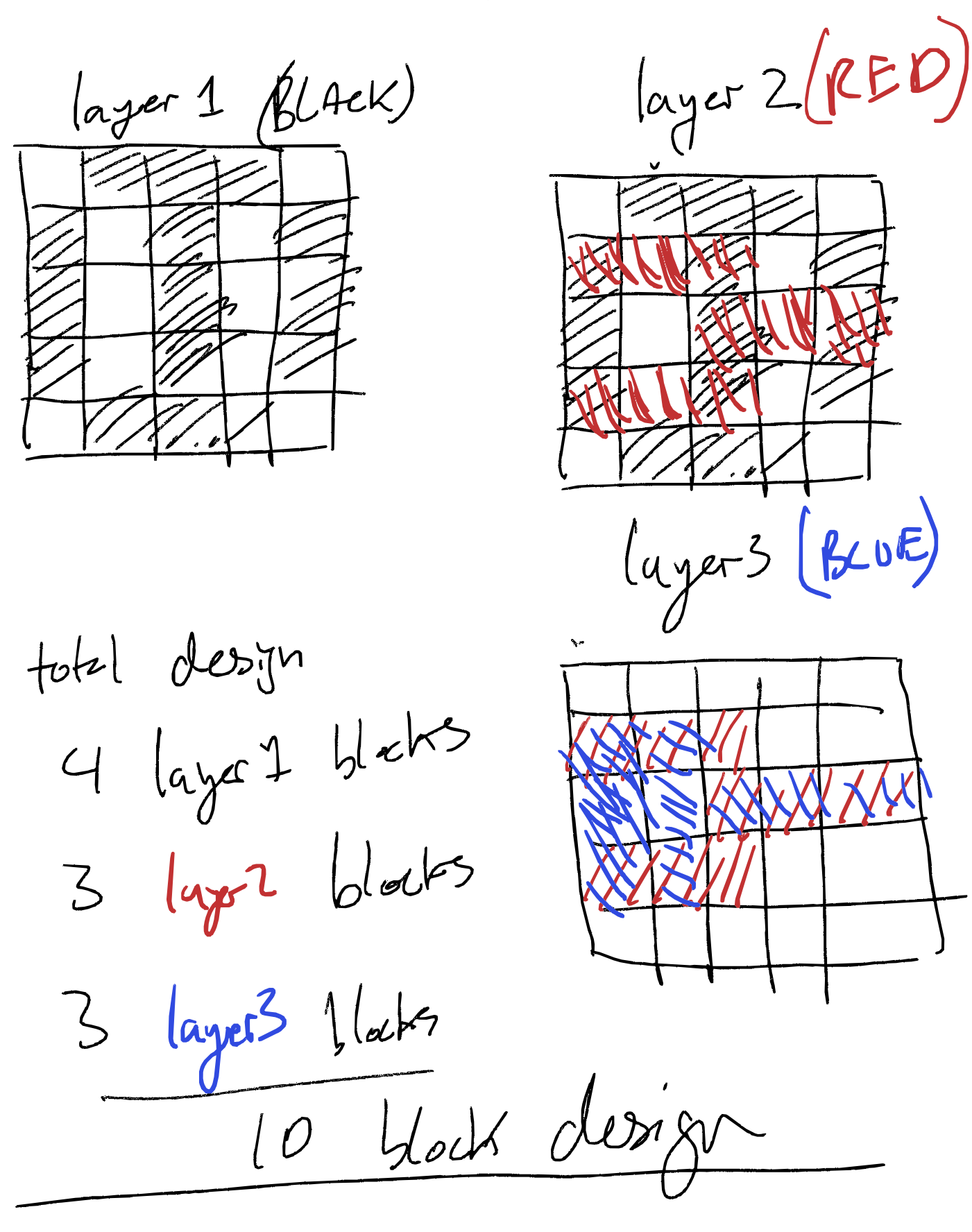
3x1

Layout area is 5x5x5



How can these be procedurally generated?

random placement of blocks on each layer
↳ move to t next layer and add blocks (dis...) such that under gravity the structure is stable



10 block design

Instructions for CLAUDE.

I am interested in designing a Jenga style modular design task for Mani-Skill. In this task IS 3x1x1 blocks (leave unit undefined for now) will lie randomly spread out on a table top surface. A 5x5x5 design area and target design will be specified. The goal of this task will be to get the manipulator to stack blocks within the design area to match the target design as closely as possible. In the following sections, I will specify how designs will be generated and scored, and how the initial table should be laid out.

GENERATING DESIGNS

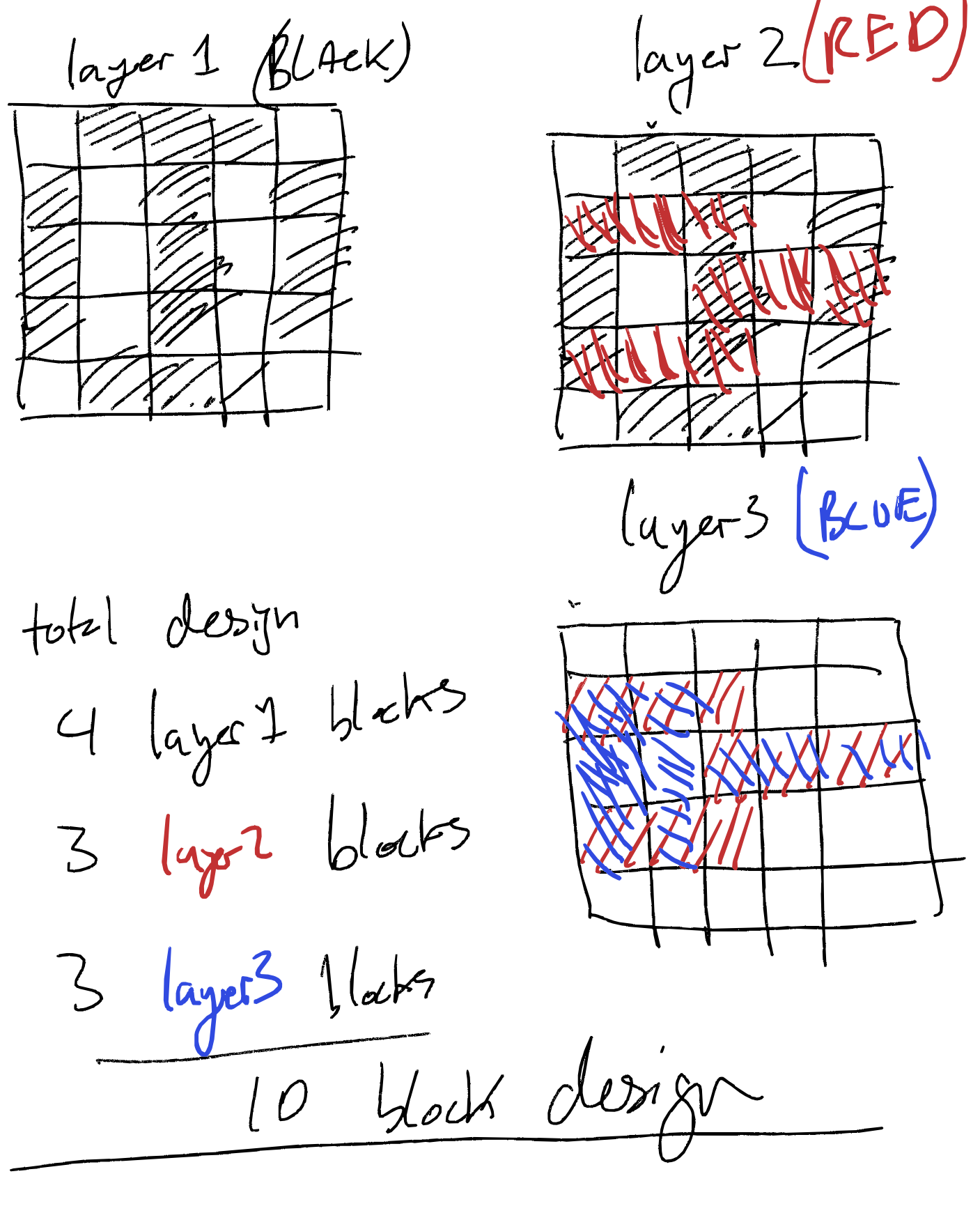
Designs will be specified as a 5x5x5 matrix of integers. Each 3x1x1 block that shows up in the design will be represented by its own integer [1 ... 41] to acknowledge the up to 41 blocks that could show up. (Most designs will be simpler w/ < 15 blocks, but # blocks allows varying difficulty). The 0 integer will indicate no block being placed in that location.

An important condition on this structure is that it is stable under gravity / physics of the Mani-Skill simulator. (NOTE: a further constraint would be that there exists a path of blocks (b_t) such that each substructure is also stable yielding a clean "construction-plan" for the robot. This should be considered, and discussed before implementation.)

Given these constraints an LLM agent or random combinatorial algorithm can come up w/ designs.

Blocks may be layed out either horizontally (3x1 side down) in either direction or vertically (1x1) side down.

Below will be an example of a design where all blocks are laid horizontally and only three layers are used (neither must be the case)



10 block design

SCORING

The result of the robot should be scored as the maximum # of blocks that occupy the same 5x5x5 grid location as blocks in the design.

I am not sure what the exact algorithm to compute this should be

INITIAL LAYOUT

Half the table should be dedicated to blocks that are randomly layed out randomly. The other half should have a 5x5 grid on it as a visual marker of the base of the design space.

The manipulator should start somewhere in between.